Custom Response - HTML, Stream, File, others

By default, **FastAPI** will return JSON responses.

You can override it by returning a `Response` directly as seen in Return a Response directly.

But if you return a `Response` directly (or any subclass, like `JSONResponse`), the data won't be automatically converted (even if you declare a `response_model`), and the documentation won't be automatically generated (for example, including the specific "media type", in the HTTP header `Content-Type` as part of the generated OpenAPI).

But you can also declare the `Response` that you want to be used (e.g. any `Response` subclass), in the *path operation decorator* using the `response_class` parameter.

The contents that you return from your *path operation function* will be put inside of that `Response`.

> **Note**
>
> If you use a response class with no media type, FastAPI will expect your response to have no content, so it will not document the response format in its generated OpenAPI docs.

JSON Responses

By default FastAPI returns JSON responses.

If you declare a Response Model FastAPI will use it to serialize the data to JSON, using Pydantic.

If you don't declare a response model, FastAPI will use the `jsonable_encoder` explained in JSON Compatible Encoder and put it in a `JSONResponse`.

If you declare a `response_class` with a JSON media type (`application/json`), like is the case with the `JSONResponse`, the data you return will be automatically converted (and filtered) with any Pydantic `response_model` that you declared in the *path operation decorator*. But the data won't be serialized to JSON bytes with Pydantic, instead it will be converted with the `jsonable_encoder` and then passed to the `JSONResponse` class, which will serialize it to bytes using the standard JSON library in Python.

JSON Performance

In short, if you want the maximum performance, use a Response Model and don't declare a `response_class` in the *path operation decorator*.

Python 3.10+

```python
# Code above omitted 👆

@app.post("/items/")
async def create_item(item: Item) -> Item:
    return item

# Code below omitted 👇
```

Ask AI

👀 **Full file preview**

Python 3.10+

```python
from fastapi import FastAPI
from pydantic import BaseModel

app = FastAPI()


class Item(BaseModel):
    name: str
    description: str | None = None
    price: float
    tax: float | None = None
    tags: list[str] = []


@app.post("/items/")
async def create_item(item: Item) -> Item:
    return item


@app.get("/items/")
async def read_items() -> list[Item]:
    return [
        Item(name="Portal Gun", price=42.0),
        Item(name="Plumbus", price=32.0),
    ]
```

HTML Response

To return a response with HTML directly from **FastAPI**, use `HTMLResponse`.

- Import `HTMLResponse`.
- Pass `HTMLResponse` as the parameter `response_class` of your *path operation decorator*.

Python 3.10+

```python
from fastapi import FastAPI
from fastapi.responses import HTMLResponse

app = FastAPI()


@app.get("/items/", response_class=HTMLResponse)
async def read_items():
    return """
    <html>
        <head>
            <title>Some HTML in here</title>
        </head>
        <body>
            <h1>Look ma! HTML!</h1>
        </body>
    </html>
    """
```

Info

The parameter `response_class` will also be used to define the "media type" of the response.

In this case, the HTTP header `Content-Type` will be set to `text/html`.

And it will be documented as such in OpenAPI.

Return a `Response`

As seen in Return a Response directly, you can also override the response directly in your *path operation*, by returning it.

The same example from above, returning an `HTMLResponse`, could look like:

Python 3.10+

```python
from fastapi import FastAPI
from fastapi.responses import HTMLResponse

app = FastAPI()


@app.get("/items/")
async def read_items():
    html_content = """
    <html>
        <head>
            <title>Some HTML in here</title>
        </head>
        <body>
            <h1>Look ma! HTML!</h1>
        </body>
    </html>
    """
    return HTMLResponse(content=html_content, status_code=200)
```

Warning

A `Response` returned directly by your *path operation function* won't be documented in OpenAPI (for example, the `Content-Type` won't be documented) and won't be visible in the automatic interactive docs.

Info

Of course, the actual `Content-Type` header, status code, etc, will come from the `Response` object you returned.

Document in OpenAPI and override `Response`

If you want to override the response from inside of the function but at the same time document the "media type" in OpenAPI, you can use the `response_class` parameter AND return a `Response` object.

The `response_class` will then be used only to document the OpenAPI *path operation*, but your `Response` will be used as is.

Return an `HTMLResponse` directly

For example, it could be something like:

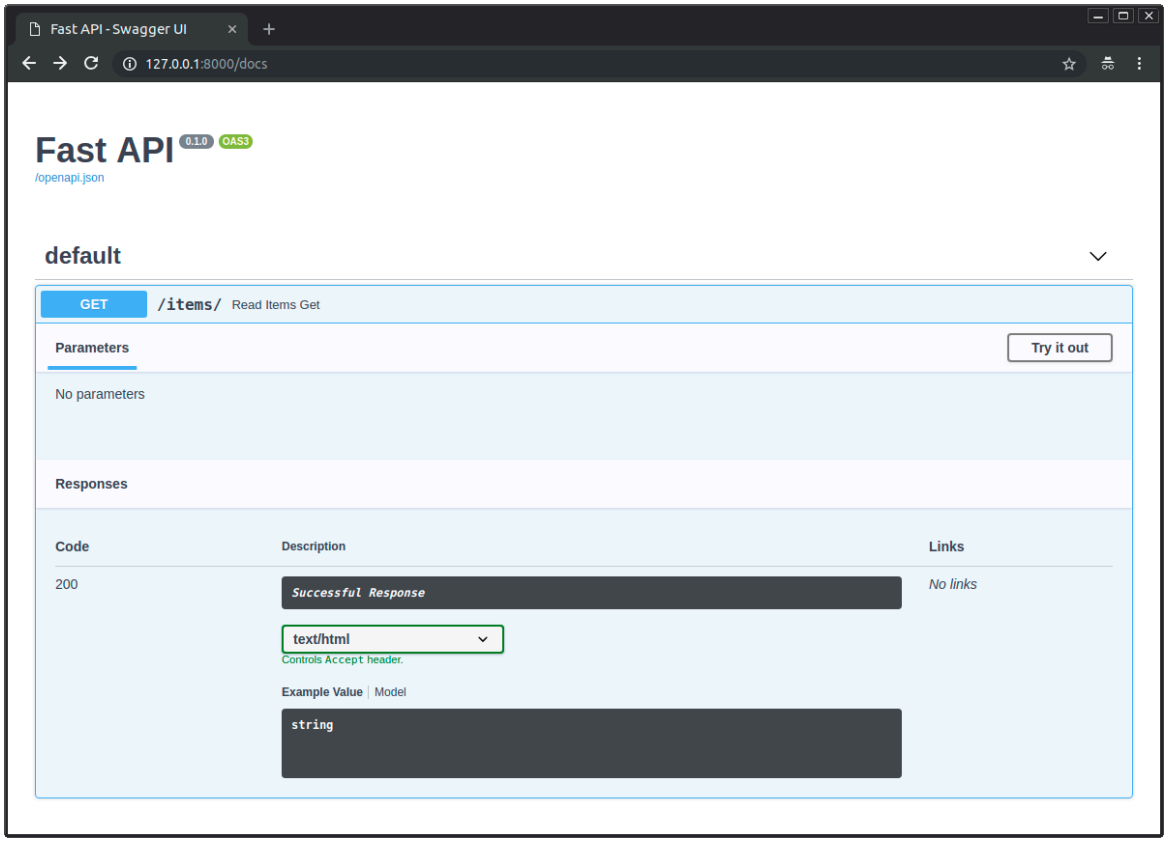
Python 3.10+

```python
from fastapi import FastAPI
from fastapi.responses import HTMLResponse

app = FastAPI()


def generate_html_response():
    html_content = """
    <html>
        <head>
            <title>Some HTML in here</title>
        </head>
        <body>
            <h1>Look ma! HTML!</h1>
        </body>
    </html>
    """
    return HTMLResponse(content=html_content, status_code=200)


@app.get("/items/", response_class=HTMLResponse)
async def read_items():
    return generate_html_response()
```

In this example, the function `generate_html_response()` already generates and returns a `Response` instead of returning the HTML in a `str`.

By returning the result of calling `generate_html_response()`, you are already returning a `Response` that will override the default **FastAPI** behavior.

But as you passed the `HTMLResponse` in the `response_class` too, **FastAPI** will know how to document it in OpenAPI and the interactive docs as HTML with `text/html`:



Available responses

Here are some of the available responses.

Keep in mind that you can use `Response` to return anything else, or even create a custom sub-class.

> **Technical Details**
>
> You could also use `from starlette.responses import HTMLResponse`.
>
> **FastAPI** provides the same `starlette.responses` as `fastapi.responses` just as a convenience for you, the developer. But most of the available responses come directly from Starlette.

`Response`

The main `Response` class, all the other responses inherit from it.

You can return it directly.

It accepts the following parameters:

- `content` - A `str` or `bytes`.
- `status_code` - An `int` HTTP status code.
- `headers` - A `dict` of strings.
- `media_type` - A `str` giving the media type. E.g. `"text/html"`.

FastAPI (actually Starlette) will automatically include a Content-Length header. It will also include a Content-Type header, based on the `media_type` and appending a charset for text types.

Python 3.10+

```python
from fastapi import FastAPI, Response

app = FastAPI()


@app.get("/legacy/")
def get_legacy_data():
    data = """<?xml version="1.0"?>
    <shampoo>
    <Header>
        Apply shampoo here.
    </Header>
    <Body>
        You'll have to use soap here.
    </Body>
    </shampoo>
    """
    return Response(content=data, media_type="application/xml")
```

`HTMLResponse`

Takes some text or bytes and returns an HTML response, as you read above.

`PlainTextResponse`

Takes some text or bytes and returns a plain text response.

Python 3.10+

```python
from fastapi import FastAPI
from fastapi.responses import PlainTextResponse

app = FastAPI()


@app.get("/", response_class=PlainTextResponse)
async def main():
    return "Hello World"
```

`JSONResponse`

Takes some data and returns an `application/json` encoded response.

This is the default response used in **FastAPI**, as you read above.

> **Technical Details**
>
> But if you declare a response model or return type, that will be used directly to serialize the data to JSON, and a response with the right media type for JSON will be returned directly, without using the `JSONResponse` class.
>
> This is the ideal way to get the best performance.

`RedirectResponse`

Returns an HTTP redirect. Uses a `307` status code (Temporary Redirect) by default.

You can return a `RedirectResponse` directly:

Python 3.10+

```python
from fastapi import FastAPI
from fastapi.responses import RedirectResponse

app = FastAPI()


@app.get("/typer")
async def redirect_typer():
    return RedirectResponse("https://typer.tiangolo.com")
```

Or you can use it in the `response_class` parameter:

Python 3.10+

```python
from fastapi import FastAPI
from fastapi.responses import RedirectResponse

app = FastAPI()


@app.get("/fastapi", response_class=RedirectResponse)
async def redirect_fastapi():
    return "https://fastapi.tiangolo.com"
```

If you do that, then you can return the URL directly from your *path operation* function.

In this case, the `status_code` used will be the default one for the `RedirectResponse`, which is `307`.

You can also use the `status_code` parameter combined with the `response_class` parameter:

Python 3.10+

```python
from fastapi import FastAPI
from fastapi.responses import RedirectResponse

app = FastAPI()


@app.get("/pydantic", response_class=RedirectResponse, status_code=302)
async def redirect_pydantic():
    return "https://docs.pydantic.dev/"
```

`StreamingResponse`

Takes an async generator or a normal generator/iterator (a function with `yield`) and streams the response body.

Python 3.10+

```python
import anyio
from fastapi import FastAPI
from fastapi.responses import StreamingResponse

app = FastAPI()


async def fake_video_streamer():
    for i in range(10):
        yield b"some fake video bytes"
        await anyio.sleep(0)


@app.get("/")
async def main():
    return StreamingResponse(fake_video_streamer())
```

Technical Details

An `async` task can only be cancelled when it reaches an `await`. If there is no `await`, the generator (function with `yield`) can not be cancelled properly and may keep running even after cancellation is requested.

Since this small example does not need any `await` statements, we add an `await anyio.sleep(0)` to give the event loop a chance to handle cancellation.

This would be even more important with large or infinite streams.

Tip

Instead of returning a `StreamingResponse` directly, you should probably follow the style in Stream Data, it's much more convenient and handles cancellation behind the scenes for you.

If you are streaming JSON Lines, follow the Stream JSON Lines tutorial.

`FileResponse`

Asynchronously streams a file as the response.

Takes a different set of arguments to instantiate than the other response types:

- `path` - The file path to the file to stream.
- `headers` - Any custom headers to include, as a dictionary.
- `media_type` - A string giving the media type. If unset, the filename or path will be used to infer a media type.
- `filename` - If set, this will be included in the response `Content-Disposition`.

File responses will include appropriate `Content-Length`, `Last-Modified` and `ETag` headers.

Python 3.10+

```python
from fastapi import FastAPI
from fastapi.responses import FileResponse

some_file_path = "large-video-file.mp4"
app = FastAPI()


@app.get("/")
async def main():
    return FileResponse(some_file_path)
```

You can also use the `response_class` parameter:

Python 3.10+

```python
from fastapi import FastAPI
from fastapi.responses import FileResponse

some_file_path = "large-video-file.mp4"
app = FastAPI()


@app.get("/", response_class=FileResponse)
async def main():
    return some_file_path
```

In this case, you can return the file path directly from your *path operation* function.

Custom response class

You can create your own custom response class, inheriting from `Response` and using it.

For example, let's say that you want to use `orjson` with some settings.

Let's say you want it to return indented and formatted JSON, so you want to use the orjson option `orjson.OPT_INDENT_2`.

You could create a `CustomORJSONResponse`. The main thing you have to do is create a `Response.render(content)` method that returns the content as `bytes`:

Python 3.10+

```python
from typing import Any

import orjson
from fastapi import FastAPI, Response

app = FastAPI()
```

```python
class CustomORJSONResponse(Response):
    media_type = "application/json"

    def render(self, content: Any) -> bytes:
        assert orjson is not None, "orjson must be installed"
        return orjson.dumps(content, option=orjson.OPT_INDENT_2)


@app.get("/", response_class=CustomORJSONResponse)
async def main():
    return {"message": "Hello World"}
```

Now instead of returning:

```
{"message": "Hello World"}
```

...this response will return:

```
{
  "message": "Hello World"
}
```

Of course, you will probably find much better ways to take advantage of this than formatting JSON. 😉

`orjson` or Response Model

If what you are looking for is performance, you are probably better off using a Response Model than an `orjson` response.

With a response model, FastAPI will use Pydantic to serialize the data to JSON, without using intermediate steps, like converting it with `jsonable_encoder`, which would happen in any other case.

And under the hood, Pydantic uses the same underlying Rust mechanisms as `orjson` to serialize to JSON, so you will already get the best performance with a response model.

Default response class

When creating a **FastAPI** class instance or an `APIRouter` you can specify which response class to use by default.

The parameter that defines this is `default_response_class`.

In the example below, **FastAPI** will use `HTMLResponse` by default, in all *path operations*, instead of JSON.

Python 3.10+

```python
from fastapi import FastAPI
from fastapi.responses import HTMLResponse

app = FastAPI(default_response_class=HTMLResponse)


@app.get("/items/")
async def read_items():
    return "<h1>Items</h1><p>This is a list of items.</p>"
```

> **Tip**
>
> You can still override `response_class` in *path operations* as before.

Additional documentation

You can also declare the media type and many other details in OpenAPI using `responses`: Additional Responses in OpenAPI.